

Lecture 9

local truncation error of single-step methods
global truncation error, consistency and convergence
higher-order methods
Richardson error estimates

local truncation error

Definition: local truncation error

Let $y_n = y(t_n)$ be given and $y_{n+1} = y(t_{n+1})$ be the exact solution of the initial-value problem at t_{n+1} and z_{n+1} its numerical approximation starting with the known $z_n = y_n$. Then, the *local truncation error* τ_{n+1} is defined as:

$$\tau_{n+1} = \frac{y_{n+1} - z_{n+1}}{\Delta t},$$

where $\Delta t = t_{n+1} - t_n$.

This is the error over a single step of a numerical method with exact initial data, relative to the step length Δt .

Local truncation error for the general case $y' = f(y, t)$

For simple, explicit numerical time-integration methods (Forward Euler, Modified Euler), it is possible to derive the order of τ_{n+1} with a general $f(y, t)$ by expanding y_{n+1} and z_{n+1} into Taylor series

Local truncation error of the Forward Euler method

$$\begin{aligned}y' &= f(t, y), \quad t_n \leq t \leq t_{n+1}, \\y(t_n) &= y_n\end{aligned}$$

Forward Euler method proceeds as:

$$w_{n+1} = w_n + \Delta t f(t_n, w_n), \quad n = 0, 1, \dots$$

Let $w_n = y_n$, then:

$$z_{n+1} = y_n + \Delta t f(t_n, y_n)$$

For the exact y_{n+1} we use the Taylor series:

$$y_{n+1} = y_n + y'_n \Delta t + \frac{y''(\xi)}{2} (\Delta t)^2 = y_n + f(t_n, y_n) \Delta t + \frac{y''(\xi)}{2} (\Delta t)^2, \quad \xi \in (t_n, t_{n+1})$$

The local truncation error is:

$$\tau_{n+1} = \frac{y_{n+1} - z_{n+1}}{\Delta t} = \frac{y''(\xi)}{2} \Delta t, \quad \xi \in (t_n, t_{n+1}).$$

Local truncation error of the Modified Euler method

Modified Euler method proceeds as:

$$\bar{w}_{n+1} = w_n + \Delta t f(t_n, w_n),$$

$$w_{n+1} = w_n + \frac{\Delta t}{2} f(t_n, w_n) + \frac{\Delta t}{2} f(t_{n+1}, \bar{w}_{n+1})$$

Let $w_n = y_n$, then:

$$\bar{z}_{n+1} = y_n + \Delta t f(t_n, y_n),$$

$$z_{n+1} = y_n + \frac{\Delta t}{2} f(t_n, y_n) + \frac{\Delta t}{2} f(t_{n+1}, \bar{z}_{n+1})$$

For the exact y_{n+1} we use the Taylor series with one extra term:

$$y_{n+1} = y_n + y'_n \Delta t + \frac{y''_n}{2} (\Delta t)^2 + \mathcal{O}((\Delta t)^3)$$

and the facts that

$$y'_n = f(t_n, y_n),$$

$$y''_n = \left. \frac{d}{dt} f(t, y) \right|_{t=t_n} = \left. \frac{\partial f}{\partial t} \frac{dt}{dt} \right|_{t=t_n} + \left. \frac{\partial f}{\partial y} \frac{dy}{dt} \right|_{t=t_n} = \left. \frac{\partial f}{\partial t} \right|_{t=t_n} + f(t_n, y_n) \left. \frac{\partial f}{\partial y} \right|_{t=t_n}$$

Local truncation error of the Modified Euler method

For the numerical solution we expand $f(t_{n+1}, \bar{z}_{n+1})$ in the Taylor series over both variables, denoting $f_n = f(t_n, y_n)$:

$$\begin{aligned} f(t_{n+1}, \bar{z}_{n+1}) &= f(t_n + \Delta t, y_n + \Delta t f(t_n, y_n)) = f(t_n + \Delta t, y_n + \Delta t f_n) \\ &= f_n + \left. \frac{\partial f}{\partial t} \right|_{t=t_n} \Delta t + \left. \frac{\partial f}{\partial y} \right|_{y=y_n} \Delta t f_n + \frac{1}{2} \left. \frac{\partial^2 f}{\partial t^2} \right|_{t=t_n} (\Delta t)^2 \\ &\quad + \frac{1}{2} \left. \frac{\partial^2 f}{\partial y^2} \right|_{y=y_n} (\Delta t f_n)^2 + \left. \frac{\partial^2 f}{\partial y \partial t} \right|_{y=y_n, t=t_n} (\Delta t)^2 f_n + \mathcal{O}((\Delta t)^3) \\ &= f_n + \left. \frac{\partial f}{\partial t} \right|_{t=t_n} \Delta t + \left. \frac{\partial f}{\partial y} \right|_{y=y_n} \Delta t f_n + \mathcal{O}((\Delta t)^2) \\ &= f(t_n, y_n) + \Delta t \left(\left. \frac{\partial f}{\partial t} \right|_{t=t_n} + f(t_n, y_n) \left. \frac{\partial f}{\partial y} \right|_{t=t_n} \right) + \mathcal{O}((\Delta t)^2) \end{aligned}$$

Local truncation error of the Modified Euler method

Hence, numerical solution z_{n+1} produced by the Modified Euler method can be written as:

$$\begin{aligned} z_{n+1} &= y_n + \frac{\Delta t}{2} f(t_n, y_n) + \frac{\Delta t}{2} f(t_{n+1}, \bar{z}_{n+1}) \\ &= y_n + \frac{\Delta t}{2} f(t_n, y_n) \\ &\quad + \frac{\Delta t}{2} \left(f(t_n, y_n) + \Delta t \left(\left. \frac{\partial f}{\partial t} \right|_{t=t_n} + f(t_n, y_n) \left. \frac{\partial f}{\partial y} \right|_{t=t_n} \right) + \mathcal{O}((\Delta t)^2) \right) \\ &= y_n + f(t_n, y_n) \Delta t + \frac{1}{2} \left(\left. \frac{\partial f}{\partial t} \right|_{t=t_n} + f(t_n, y_n) \left. \frac{\partial f}{\partial y} \right|_{t=t_n} \right) (\Delta t)^2 + \mathcal{O}((\Delta t)^3) \end{aligned}$$

and the exact solution y_{n+1} is

$$y_{n+1} = y_n + f(t_n, y_n) \Delta t + \frac{1}{2} \left(\left. \frac{\partial f}{\partial t} \right|_{t=t_n} + f(t_n, y_n) \left. \frac{\partial f}{\partial y} \right|_{t=t_n} \right) (\Delta t)^2 + \mathcal{O}((\Delta t)^3)$$

Local truncation error of the Modified Euler method

Therefore, the local error is:

$$\tau_{n+1} = \frac{y_{n+1} - z_{n+1}}{\Delta t} = \frac{\mathcal{O}((\Delta t)^3) - \mathcal{O}((\Delta t)^3)}{\Delta t} = \frac{\mathcal{O}((\Delta t)^3)}{\Delta t} = \mathcal{O}((\Delta t)^2)$$

Hence, the Forward Euler method is $\mathcal{O}(\Delta t)$, whereas the Modified Euler method is $\mathcal{O}((\Delta t)^2)$.

Local truncation error for the test equation $y' = \lambda y$

For any single-step numerical time-integration method with a known amplification factor Q , it is possible to derive the order of τ_{n+1} with $f(y, t) = \lambda y$ by expanding y_{n+1} and Q in Taylor series

Local truncation error for a test equation

$$\begin{aligned}y' &= \lambda y, \quad t_n \leq t \leq t_{n+1}, \\y(t_n) &= y_n\end{aligned}$$

The exact solution at t_{n+1} is

$$y_{n+1} = y_n e^{\lambda \Delta t}$$

Any single-step numerical method, including implicit ones, can be written as:

$$w_{n+1} = Q w_n, \quad n = 0, 1, \dots$$

where $Q(\lambda \Delta t)$ is the amplification factor of the method. Let $w_n = y_n$, then the numerical solution is:

$$z_{n+1} = Q y_n$$

and the local truncation error is:

$$\tau_{n+1} = \frac{y_{n+1} - z_{n+1}}{\Delta t} = \frac{y_n e^{\lambda \Delta t} - Q y_n}{\Delta t} = \frac{e^{\lambda \Delta t} - Q}{\Delta t} y_n$$

Local truncation error for a test equation

Expanding the exponent in its Taylor series:

$$e^{\lambda \Delta t} = 1 + \lambda \Delta t + \frac{1}{2}(\lambda \Delta t)^2 + \dots = \sum_{k=0}^{\infty} \frac{1}{k!}(\lambda \Delta t)^k$$

we get

$$\tau_{n+1} = \frac{e^{\lambda \Delta t} - Q}{\Delta t} y_n = \frac{y_n}{\Delta t} \left(\sum_{k=0}^{\infty} \frac{1}{k!}(\lambda \Delta t)^k - Q \right)$$

This general expression was derived for the test equation with $f(t, y) = \lambda y$ and is valid for any Q , i.e., for any single-step numerical method.

Local truncation error of Backward Euler method

Amplification factor of the Backward Euler method is:

$$Q = \frac{1}{1 - \lambda \Delta t}$$

Expanding it as the Taylor series:

$$\frac{1}{1 - \lambda \Delta t} = 1 + \lambda \Delta t + (\lambda \Delta t)^2 + \dots = \sum_{k=0}^{\infty} (\lambda \Delta t)^k$$

we get (how/why?)

$$\begin{aligned} \tau_{n+1} &= \frac{y_n}{\Delta t} \left(\sum_{k=0}^{\infty} \frac{1}{k!} (\lambda \Delta t)^k - (\lambda \Delta t)^k \right) = \frac{y_n}{\Delta t} \left(\sum_{k=2}^{\infty} \frac{1}{k!} (\lambda \Delta t)^k - (\lambda \Delta t)^k \right) \\ &= \mathcal{O}(\Delta t) \end{aligned}$$

The Backward Euler method is $\mathcal{O}(\Delta t)$ for the test equation.

Convergence of the method and its global truncation error

To investigate convergence and derive the order of the global truncation error, it is sufficient to know the stability properties of the method and its local truncation error τ_{n+1}

Definition: global truncation error

The *global truncation error* e_{n+1} is defined as the difference between the exact solution y_{n+1} at t_{n+1} and its numerical approximation w_{n+1} :

$$e_{n+1} = y_{n+1} - w_{n+1},$$

where $w_0 = y_0$.

This is the error at a fixed time instant, which could be many steps away from t_0 .

Definition: consistency

A numerical time-integration method is called *consistent* if

$$\lim_{\Delta t \rightarrow 0} \tau_{n+1} = 0,$$

where τ_{n+1} is the local truncation error. A method is called *consistent of order p* if

$$\tau_{n+1} = \mathcal{O}((\Delta t)^p)$$

Definition: convergence

A numerical time-integration method is called *convergent* if

$$\lim_{\Delta t \rightarrow 0} e_{n+1} = 0,$$

where $T = (n + 1)\Delta t$ is a fixed time instant and e_{n+1} is the global truncation error. A method is called *convergent of order p* if

$$e_{n+1} = \mathcal{O}((\Delta t)^p)$$

A convergent method means that the numerical solution converges to the exact solution as $\Delta t \rightarrow 0$.

Theorem (Lax): equivalence

If a numerical time-integration method is stable and consistent, then the method is also convergent. Moreover, the global and local truncation errors have the same order.

Proof for the test equation is in the book.

Studying convergence of a method

Studying the convergence of a numerical time-integration method means establishing the conditions on Δt when the method converges and the rate at which it converges (order of convergence). You should be able to do it for the test equation as follows:

- 1 Derive the amplification factor Q of the method.
- 2 Use Q to derive the stability condition of the method in the form:
 $0 < \Delta t \leq M$. For unconditionally stable methods: $0 < \Delta t < \infty$.
- 3 Use Q (or its Taylor series) and the test equation to derive the order p of the local truncation error, $\tau_{n+1} = \mathcal{O}((\Delta t)^p)$.
- 4 Use Lax Equivalence Theorem to conclude that, for $0 < \Delta t \leq M$, the method is convergent with order p , i.e., $e_{n+1} = \mathcal{O}((\Delta t)^p)$.

Studying convergence of a higher-order Runge-Kutta method

Runge-Kutta (RK4) method

The RK4 method proceeds as follows:

$$\begin{aligned}k_1 &= \Delta t f(t_n, w_n), \\k_2 &= \Delta t f\left(t_n + \frac{1}{2}\Delta t, w_n + \frac{1}{2}k_1\right), \\k_3 &= \Delta t f\left(t_n + \frac{1}{2}\Delta t, w_n + \frac{1}{2}k_2\right), \\k_4 &= \Delta t f(t_n + \Delta t, w_n + k_3), \\w_{n+1} &= w_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4).\end{aligned}$$

Amplification factor of the RK4 method

To compute the Q -factor, we can simply apply the method to the test equation:

$$y' = \lambda y$$

with $f(t, y) = \lambda y$ as follows:

$$k_1 = \Delta t \lambda w_n,$$

$$k_2 = \Delta t \lambda \left(w_n + \frac{1}{2} \Delta t \lambda w_n \right) = \Delta t \lambda w_n + \frac{1}{2} (\Delta t \lambda)^2 w_n,$$

$$\begin{aligned} k_3 &= \Delta t \lambda \left(w_n + \frac{1}{2} \left(\Delta t \lambda w_n + \frac{1}{2} (\Delta t \lambda)^2 w_n \right) \right) \\ &= \Delta t \lambda w_n + \frac{1}{2} (\Delta t \lambda)^2 w_n + \frac{1}{4} (\Delta t \lambda)^3 w_n, \end{aligned}$$

$$\begin{aligned} k_4 &= \Delta t \lambda \left(w_n + \Delta t \lambda w_n + \frac{1}{2} (\Delta t \lambda)^2 w_n + \frac{1}{4} (\Delta t \lambda)^3 w_n \right) \\ &= \Delta t \lambda w_n + (\Delta t \lambda)^2 w_n + \frac{1}{2} (\Delta t \lambda)^3 w_n + \frac{1}{4} (\Delta t \lambda)^4 w_n \end{aligned}$$

Amplification factor of the RK4 method

$$\begin{aligned}w_{n+1} &= w_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \\&= w_n + \frac{w_n}{6} \left[\Delta t \lambda + 2 \left(\Delta t \lambda + \frac{1}{2}(\Delta t \lambda)^2 \right) \right. \\&\quad \left. + 2 \left(\Delta t \lambda + \frac{1}{2}(\Delta t \lambda)^2 + \frac{1}{4}(\Delta t \lambda)^3 \right) \right. \\&\quad \left. + \Delta t \lambda + (\Delta t \lambda)^2 + \frac{1}{2}(\Delta t \lambda)^3 + \frac{1}{4}(\Delta t \lambda)^4 \right] \\&= w_n + \frac{w_n}{6} \left[6\Delta t \lambda + 3(\Delta t \lambda)^2 + (\Delta t \lambda)^3 + \frac{1}{4}(\Delta t \lambda)^4 \right] \\&= \left[1 + \Delta t \lambda + \frac{1}{2}(\Delta t \lambda)^2 + \frac{1}{6}(\Delta t \lambda)^3 + \frac{1}{24}(\Delta t \lambda)^4 \right] w_n\end{aligned}$$

Hence, the amplification factor of RK4 is

$$Q = 1 + \Delta t \lambda + \frac{1}{2}(\Delta t \lambda)^2 + \frac{1}{6}(\Delta t \lambda)^3 + \frac{1}{24}(\Delta t \lambda)^4$$

Stability of the RK4 method

We assume that the test problem is stable, i.e., $\lambda = -|\lambda| < 0$. RK4 is stable if $|Q| \leq 1$. Denoting $x = \Delta t \lambda$, this can be written as

$$\left| 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 \right| \leq 1$$
$$-1 \leq 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 \leq 1$$

Left inequality is always satisfied. Considering the right inequality we get:

$$x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 \leq 0$$
$$x \left(1 + \frac{1}{2}x + \frac{1}{6}x^2 + \frac{1}{24}x^3 \right) \leq 0$$
$$1 + \frac{1}{2}x + \frac{1}{6}x^2 + \frac{1}{24}x^3 \geq 0$$

since $x = -|\lambda|\Delta t < 0$ for $\Delta t > 0$.

Stability of the RK4 method

Looking separately at the polynomial

$$P(x) = 1 + \frac{1}{2}x + \frac{1}{6}x^2 + \frac{1}{24}x^3$$

we can show that the sign of $P(x)$ changes around $x \approx -2.8$:

$$\begin{cases} P(x) < 0, & x < -2.8, \\ P(x) = 0, & x \approx -2.8, \\ P(x) > 0, & x > -2.8 \end{cases}$$

Hence, the approximate stability condition can be derived from $x \geq -2.8$:

$$\begin{aligned} -|\lambda|\Delta t &\geq -2.8 \\ 0 < \Delta t &\leq \frac{2.8}{|\lambda|} \end{aligned}$$

Is it better or worse than the stability condition for the Forward Euler method?

Local truncation error of the RK4 method

The error is established for the test equation:

$$\begin{aligned}\tau_{n+1} &= \frac{y_n}{\Delta t} \left(\sum_{k=0}^{\infty} \frac{1}{k!} (\lambda \Delta t)^k - Q \right) \\ &= \frac{y_n}{\Delta t} \left(\sum_{k=0}^{\infty} \frac{1}{k!} (\lambda \Delta t)^k - 1 - \Delta t \lambda - \frac{1}{2} (\Delta t \lambda)^2 - \frac{1}{6} (\Delta t \lambda)^3 - \frac{1}{24} (\Delta t \lambda)^4 \right) \\ &= \frac{y_n}{\Delta t} \mathcal{O}((\lambda \Delta t)^5) = \mathcal{O}((\Delta t)^4)\end{aligned}$$

Global truncation error of the RK4 method

- 1 The method is stable for $0 < \Delta t \leq 2.8/|\lambda|$
- 2 The local truncation error $\tau_n = \mathcal{O}((\Delta t)^4)$
- 3 Since $\lim_{\Delta t \rightarrow 0} \tau_n = 0$, the method is consistent of order $p = 4$.
- 4 Hence, by the Lax Equivalence Theorem, for $0 < \Delta t \leq 2.8/|\lambda|$, the RK4 method is a convergent method of order $p = 4$.

Estimating global error by Richardson's extrapolation

Richardson's global error estimate

The main assumption is that the global error $e(t, \Delta t) = y(t) - w_n(\Delta t)$, where $t = n\Delta t$, behaves like:

$$e(t, \Delta t) = c_p(t)(\Delta t)^p + \mathcal{O}((\Delta t)^{p+1})$$

Computing the numerical solution at t with Δt , $\Delta t/2$, and $\Delta t/4$, we get:

$$\begin{aligned}y(t) - w_n(\Delta t) &= c_p(t)(\Delta t)^p, \\y(t) - w_{2n}(\Delta t/2) &= c_p(t)(\Delta t/2)^p = c_p(t)2^{-p}(\Delta t)^p, \\y(t) - w_{4n}(\Delta t/4) &= c_p(t)(\Delta t/4)^p = c_p(t)2^{-p}2^{-p}(\Delta t)^p.\end{aligned}$$

$$\begin{aligned}w_n(\Delta t) - w_{2n}(\Delta t/2) &= c_p(t)(\Delta t)^p(2^{-p} - 1), \\w_{2n}(\Delta t/2) - w_{4n}(\Delta t/4) &= c_p(t)(\Delta t)^p2^{-p}(2^{-p} - 1).\end{aligned}$$

Hence, we can now estimate both the order p and the constant c_p :

$$\begin{aligned}2^{-p} &= \frac{w_{2n}(\Delta t/2) - w_{4n}(\Delta t/4)}{w_n(\Delta t) - w_{2n}(\Delta t/2)}, \\c_p(t) &= \frac{w_n(\Delta t) - w_{2n}(\Delta t/2)}{(\Delta t)^p(2^{-p} - 1)},\end{aligned}$$

Richardson's extrapolation (improvement of the solution)

Since:

$$e(t, \Delta t) = y(t) - w_n(\Delta t) = c_p(t)(\Delta t)^p + \mathcal{O}((\Delta t)^{p+1})$$

and having estimated p and c_p , we can now extrapolate (improve) the numerical solution $w_n(t)$ as:

$$y(t) = w_n(\Delta t) + c_p(t)(\Delta t)^p + \mathcal{O}((\Delta t)^{p+1}).$$

- $w_n(t)$ is an $\mathcal{O}((\Delta t)^p)$ approximation of $y(t)$
- $w_n(t) + c_p(t)(\Delta t)^p$ is an $\mathcal{O}((\Delta t)^{p+1})$ approximation of $y(t)$

Questions

- Study the convergence of the Trapezoidal method:
 - ① what theorem and definitions are used to establish the convergence of a method?
 - ② is the method consistent?
 - ③ is the method stable? if so, for which range of Δt 's?
 - ④ does the method converge? if so, for what range of Δt 's?
 - ⑤ what is the order of convergence (order of the global error)?
- Extrapolate a numerical approximation of the solution obtained with any time-integration method to a higher-order approximation using the Richardson method
 - ① assume that $w_n(\Delta t)$ is the numerical approximation of the solution $y(t)$ of a differential equation obtained with the Trapezoidal method
 - ② what is the main assumption of Richardson's method?
 - ③ what calculation steps are required to estimate p and c_p ?
 - ④ how can one extrapolate (improve) $w_n(\Delta t)$?
 - ⑤ what is the order of the obtained extrapolated numerical approximation?